

Kaung Myat Khant

The Chi-Square Distribution & Categorical Data Analysis

2026-10-12

1. Core concepts

- **Standard Normal Origin**: Derived directly from independent standard normal variables ($z_i \sim N(0,1)$)
- **Symbol Distinction**: χ^2 denotes theoretical distribution/test; X^2 represents sample test statistic.
- **Primary Scope**: Analyzes categorical, binary, count data, and single population variances (σ^2)
- **Cardinal Rule**: Strictly requires actual raw counts of occurrences—never percentages, proportions, or averages.
- **Key Metric**: $X^2 = \sum \frac{(O-E)^2}{E}$ measures discrepancy between observed (O) and expected (E) counts.

2. Mathematical derivation & key parameters

- **Single variable relation**: Square of a standard normal score (z^2) (follows χ^2 with 1 degree of freedom).
- **Summation formula**: $\chi^2_{(k)} = \sum_{i=1}^{k} z_i^2$ for k independent standard normal variables.
- **Mean parameter**: Equal to its degrees of freedom ($\mu = k$).
- **Variance parameter**: Equal to twice its degrees of freedom ($\sigma^2 = 2k$)
- **Mode parameter**: Equal to $k - 2$ for $k \geq 2$ (0 for $k = 1$).
- **Non-Negative Domain**: Values range from 0 to $+\infty (0 \leq \chi^2 < \infty)$.
- **Asymmetric Curve**: Positively skewed with right tail; becomes symmetrical as $df \to \infty$.
- **Additive Property**: Sum of two independent chi-square variables is also a chi-square variable.

3. Scope of applications

- **Single population variance**: Inference for σ^2 via $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$ with $df = n - 1$.
- **Goodness-of-Fit tests**: Compares sample frequencies against theoretical distributions (Normal, Poisson, etc.).
- **Contingency tables ($r \times c$)**: Tests of Independence and Homogeneity with $df = (r - 1)(c - 1)$.
- 2×2 Fourfold Tables: Compares two proportions with Yates's continuity correction or Fisher's Exact Test.
- **Trend test in Proportions**: Evaluates ordered categorical data using linear scoring.
- **Paired binary outcomes**: Evaluates repeated binary measures using McNemar's test ($X^2 = \frac{(f-g)^2}{f+g}$)
- **Survival analysis**: Compares survival curves between independent groups using the log-rank test.
- **Nonparametric limit**: Asymptotic limiting distribution for Kruskal-Wallis (H) and Friedman tests (χ_r^2).

4. Step-by-step 2×2 contingency table example

Group	Passed	Failed	Total
Study Guide	15	5	20
No Guide	5	15	20
Total	20	20	40

- **Study scenario & raw counts (O)**
- **Context**: Evaluating whether a new study guide influences exam pass rates ($N = 40$ students).
- **Study Guide & Passed**: 15 students (a).
- **Study Guide & Failed**: 5 students (b).
- **No Guide & Passed**: 5 students (c).
- **No Guide & Failed**: 15 students (d).

Group	Passed	Failed	Total
Study Guide	15	5	20
No Guide	5	15	20
Total	20	20	40

Step-by-step calculation

- **Expected Count Formula**: $E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total } (N)}$.
- **Expected Cell Counts (E)**: $\frac{20 \times 20}{40} = 10$ for all four cells.
- **Guide & Passed Score**: $\frac{(15-10)^2}{10} = \frac{25}{10} = 2.5$.
- **Guide & Failed Score**: $\frac{(5-10)^2}{10} = \frac{25}{10} = 2.5$.
- **No Guide & Passed Score**: $\frac{(5-10)^2}{10} = \frac{25}{10} = 2.5$.
- **No Guide & Failed Score**: $\frac{(15-10)^2}{10} = \frac{25}{10} = 2.5$.
- **Sum of Cell Scores (X^2)**: $X^2 = 2.5 + 2.5 + 2.5 + 2.5 = 10.0$.
- 2×2 Shortcut Formula: $X^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} = \frac{40(225-25)^2}{160{,}000} = 10.0$.
- **Statistical Decision**: Calculated $X^2 = 10.0 > 3.841$ (critical value at $df = 1, \alpha = 0.05$), rejecting H_0 ($P < 0.005$).

6. R Codes

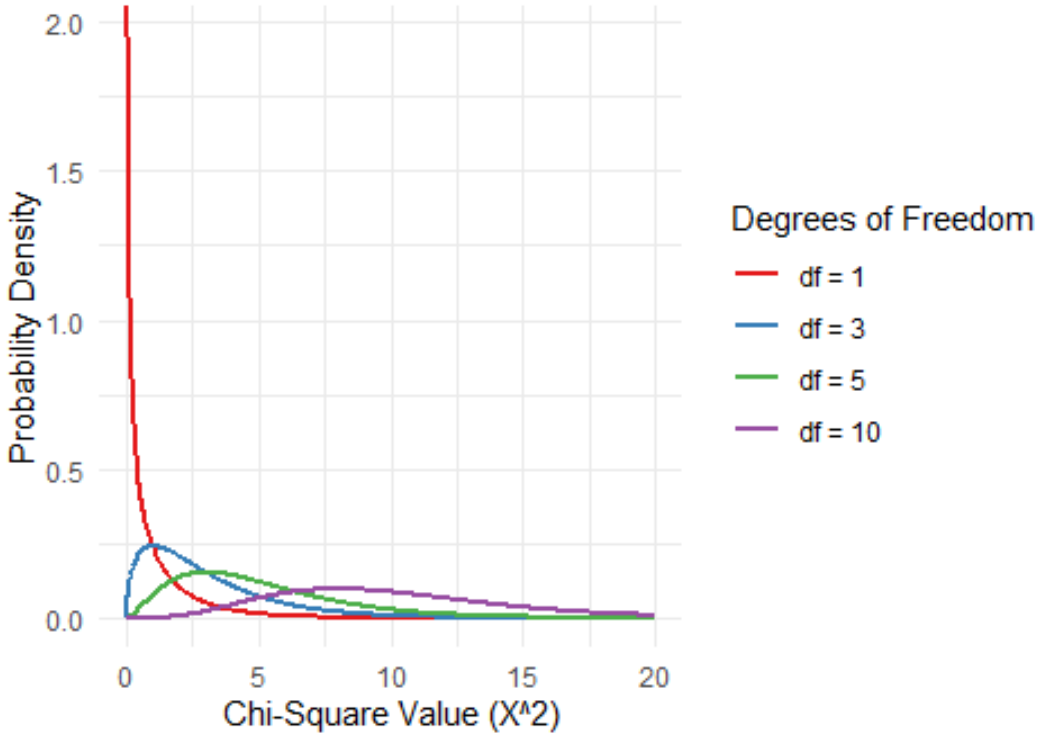
```
# --- 1. Construct 2x2 Matrix for Study Guide Example ---
guide_matrix <- matrix(c(15, 5, 5, 15),
                       nrow = 2,
                       byrow = TRUE,
                       dimnames = list(
                           Group = c("Guide", "No Guide"),
                           Result = c("Passed", "Failed")))
qchisq(p=0.95, df = 1)
# --- 2. Pearson's Chi-Square Test (Without Continuity
Correction) ---
chisq.test(guide_matrix, correct = FALSE)

# --- 3. Yates's Continuity Correction (For 2x2 Tables) ---
chisq.test(guide_matrix, correct = TRUE)

# --- 4. Fisher's Exact Test (For Small Expected Counts) ---
fisher.test(guide_matrix)

# --- 5. Multi-Cell Contingency Table Example (Campbell Suburb
Data) ---
suburb_matrix <- matrix(c(17, 25, 39, 42, 32,
                          5, 21, 34, 49, 25),
                        nrow = 5,
                        dimnames = list(Suburb = c("Manor",
                                                   "Brightside",
                                                   "Netheredge",
                                                    "Ecclesall",
                                                    "Fulwood"),
                                        Unit = c("Poisoning",
                                                 "Gastro")))
chisq.test(suburb_matrix)
```

7. Chi-Square Distribution Curves across Degrees of Freedom



8. References

1. **Daniel, Wayne W.** *Biostatistics: A Foundation for Analysis in the Health Sciences* (10th Edition). John Wiley & Sons.
2. **Campbell, Michael J.** *Statistics at Square One* (12th Edition). Wiley-Blackwell / BMJ Books.